

SMARTBRIDGE / SKILL WALLET

Project Proposal Blueprint

Date: 2 July 2026	Team ID: —
Project Name: Caption AI : AI Powered Social Media Caption Generator	

Project Proposal (Proposed Solution) Report

The proposal report outlines the transformation of social media content workflows using state-of-the-art cloud inference and local micro-servers. It aims to completely eliminate writing fatigue and platform copy formatting bottlenecks, ensuring maximum digital marketing output.

Project Overview

Component	Details & Specifications
Objective	To completely automate multi-platform social media caption copywriting cycles using fast, remote natural language processing inference nodes.
Scope	Encompasses standalone frontend web interface layout construction, integration of secure local application server micro-routers, and public routing proxy agents.
Problem Statement	Manual caption creation requires hours of exhausting work to adjust hashtags and style sheets per platform, stalling digital outreach velocity.
Impact	Drastically reduces description generation timelines from several hours to milliseconds, amplifying digital engagement and scaling operations smoothly.
Proposed Approach	Leverage Python Flask micro-servers linked to remote cloud-hosted Groq LLM inference pipelines for sub-second, intelligent copy variations processing.
Key Features	• Dynamic tone selections and automated visual prompt description processing. • Character constraint checking per platform ecosystem. • Environment isolation config layers preventing API secret vulnerabilities.

Resource Requirements

Resource Category	Description	Specification / Allocation
Hardware	Localized Sandboxed Host	Standard workstation processing core (minimum 8GB system RAM allocated)
Software Frameworks	Backend Server Architecture	Python Flask Core Routing Infrastructure
Libraries & Modules	Security & API Pipeline Helpers	python-dotenv (Environment Layer Management), groq (Cloud LLM API Client Node)
Development Space	Local Engineering Workspace	VS Code Engine Workspace Sandbox Environments
Data Management	Transient Storage Layer	Local runtime .env security keys configuration map files
Network Proxy Relay	External Routing Proxy	Ngrok Core Executable Tunnels for edge port validation routes